

Reopening Openness to Experience:

A Network Analysis of Four Openness to Experience Inventories

Supplementary Information

SI 1. Technical information about the TMFG algorithm (Massara et al., 2016)

The TMFG algorithm starts from a clique of order 4 (K_4) and adds nodes in the middle of a 3-clique (T_2 move). At each step, the algorithm optimizes the sum of the weights of the edges. T_2 is applied to the vertex and face pair that leads to the maximum increase in weights.

TMFG Algorithm:

input : A dense $n \times n$ square matrix W with positive weights

output: A sparse matrix, P , a filtered version of W fulfilling the planarity constraint

1. $C_1 \leftarrow \text{Tetrahedron}, \{v_1, v_2, v_3, v_4\}$, with highest overall total weights; // Assign the four triangular faces in C_1 to the array T
2. $T \leftarrow \{\{v_1, v_2, v_3\}, \{v_1, v_2, v_4\}, \{v_1, v_3, v_4\}, \{v_2, v_3, v_4\}\}$; // Put the $p - 4$ vertices not belonging to C_1 in the array V
3. $V \leftarrow v_5, \dots, v_p$; // Create an empty list of Separators
4. $S \leftarrow \emptyset$; // Assign the first tetrahedron to the list of cliques
5. $C \leftarrow C_1$;
6. $P \leftarrow W(C_1, C_1)$;
7. Calculate *MaxGain* for T and V as in Eq.(1) ;
8. Calculate *BestVertex* for T and V as in Eq. (2) ; // Insert $p - 4$ vertices via T_2
9. **while** V is not empty **do**
10. Find the $t_i \in T$ and $v_i \in V$ that achieve the maximum in *MaxGain* ;
11. Insert v_i into t_i // this creates three new triangles t_a, t_b, t_c
12. $V \leftarrow V \setminus v_i$;
13. $T \leftarrow (T \setminus \{t_i\}) \cup \{t_a, t_b, t_c\}$;
14. $S_i \leftarrow \{t_i\}$;
15. $S \leftarrow S \cup S_i$;
16. $C_i \leftarrow \{t_i, t_a, t_b, t_c\}$;
17. $C \leftarrow C \cup C_i$;
18. $P \leftarrow P + W(C_i, C_i) - W(S_i, S_i)$;
19. Update *MaxGain* and *BestVertex* to reflect the changes in T and V ;
20. **end**
21. **return** P ;

The resulting TMFG network is a chordal graph. A sparse inverse covariance matrix $\mathbf{J}$ that can be associated to the TMFG in such a way that edges of the TMFG correspond to non-zero values of the sparse inverse. The joint probability distribution of n variables $X_1, \dots, X_n$ is

$$P(X_1, \dots, X_n) = \frac{1}{Z} \exp\left(-\sum_{i=1}^n X_i \mathbf{J}_{i,j} X_j\right)$$

with Z a normalization constant called partition function. The elements of the inverse covariance are associated with the 4-cliques (C) and the 3-cliques separators (S) of the TMFG and can be directly computed from the covariance (see Barfuss et al., 2016, Eqs.4 and 5):

$$\mathbf{J}_{i,j} = \sum_{C \text{ s.t. } \{i,j\} \in C} (\Sigma(C)^{-1})_{i,j} - \sum_{S \text{ s.t. } \{i,j\} \in S} (\Sigma(S)^{-1})_{i,j}$$

Analogously with lasso method, $\mathbf{J}$, is a sparse inverse covariance where non-zero entries correspond to conditionally depended couples of variables.

SI 2. Openness to Experience Inventories with facet, network-identified facet, item, and gradation classification (C = core, I = intermediate, P = peripheral), and description (sorted alphabetically by inventory and then network name).

Inventory	Inventory Facet	Walktrap Facet	Network Name	Gradation	Item
BFAS	Intellect	Self-Assessed Intelligence	I10	P	I formulate ideas clearly
BFAS	Intellect	Self-Assessed Intelligence	It1	C	I have difficulty understanding abstract ideas
BFAS	Intellect	Intellectual Interests	It2	C	I avoid philosophical discussions
BFAS	Intellect	Self-Assessed Intelligence	It3	C	I avoid difficult reading material
BFAS	Intellect	Intellectual Curiosity	It4	C	I like to solve complex problems
BFAS	Intellect	Self-Assessed Intelligence	It5	I	I am quick to understand things
BFAS	Intellect	Self-Assessed Intelligence	It6	P	I can handle a lot of information
BFAS	Intellect	Self-Assessed Intelligence	It7	P	I have a rich vocabulary
BFAS	Intellect	Self-Assessed Intelligence	It8	P	I think quickly
BFAS	Intellect	Self-Assessed Intelligence	It9	C	I learn things slowly
BFAS	Openness	Fantasy	O10	I	I seldom daydream
BFAS	Openness	Aesthetic Appreciation	Op1	C	I believe in the importance of art
BFAS	Openness	Aesthetic Appreciation	Op2	I	I enjoy the beauty of nature
BFAS	Openness	Intellectual Curiosity	Op3	C	I love to reflect on things
BFAS	Openness	Aesthetic Appreciation	Op4	C	I do not like poetry
BFAS	Openness	Aesthetic Appreciation	Op5	I	I seldom notice the emotional aspects of paintings and pictures
BFAS	Openness	Imaginative	Op6	I	I need a creative outlet

BFAS	Openness	Aesthetic Appreciation	Op7	C	I see beauty in things that others might not notice
BFAS	Openness	Openness to Emotions	Op8	I	I get deeply immersed in music
BFAS	Openness	Fantasy	Op9	P	I seldom get lost in thought
HEXACO	Creativity	Imaginative	Cr1	P	I would like a job that requires following a routine rather than being creative
HEXACO	Creativity	Imaginative	Cr2	C	I would enjoy creating a work of art, such as a novel, a song, or a painting
HEXACO	Creativity	Imaginative	Cr3	C	I don't think of myself as the artistic or creative type
HEXACO	Creativity	Imaginative	Cr4	I	People have often told me that I have a good imagination
HEXACO	Aesthetic Appreciation	Aesthetic Appreciation	HA1	C	I would be quite bored by a visit to an art gallery
HEXACO	Aesthetic Appreciation	Aesthetic Appreciation	HA2	I	I wouldn't spend my time reading a book of poetry
HEXACO	Aesthetic Appreciation	Aesthetic Appreciation	HA3	I	If I had the opportunity, I would like to attend a classical music concert
HEXACO	Aesthetic Appreciation	Aesthetic Appreciation	HA4	P	Sometimes I like to just watch the wind as it blows through the trees
HEXACO	Inquisitiveness	Intellectual Interests	Iq1	I	I'm interested in learning about the history and politics of other countries
HEXACO	Inquisitiveness	Intellectual Interests	Iq2	P	I enjoy looking at maps of different places
HEXACO	Inquisitiveness	Intellectual Interests	Iq3	C	I would be very bored by a book about the history of science and technology
HEXACO	Inquisitiveness	Intellectual Interests	Iq4	C	I've never really enjoyed looking through an encyclopedia
HEXACO	Unconventionality	Intellectual Interests	Uc1	I	I think that paying attention to radical ideas is a waste of time
HEXACO	Unconventionality	Intellectual Interests	Uc2	C	I find it boring to discuss philosophy
HEXACO	Unconventionality	Imaginative	Uc3	P	I think of myself as a somewhat eccentric person
HEXACO	Unconventionality	Diversity	Uc4	P	I like people who have unconventional views
NEO	Actions	Intellectual Curiosity	Ac1	P	learn/develop new hobbies
NEO	Actions	Diversity	Ac2	P	variety/spice of life

NEO	Actions	Diversity	Ac3	P	try foreign foods
NEO	Actions	Variety-Seeking	Ac4	I	set in ways
NEO	Actions	Variety-Seeking	Ac5	P	vacation to tried and true spot
NEO	Actions	Variety-Seeking	Ac6	P	prefer time in familiar surroundings
NEO	Actions	Variety-Seeking	Ac7	I	prefer old methods
NEO	Actions	Variety-Seeking	Ac8	P	same route to go someplace
NEO	Fantasy	Imaginative	Fa1	C	active imagination
NEO	Fantasy	Imaginative	Fa2	I	active fantasy
NEO	Fantasy	Imaginative	Fa3	I	letting a fantasy or daydream develop
NEO	Fantasy	Fantasy	Fa4	I	don't waste time daydreaming
NEO	Fantasy	Fantasy	Fa5	P	keep thoughts on realistic lines
NEO	Fantasy	Fantasy	Fa6	P	difficulty letting my mind wander
NEO	Fantasy	Fantasy	Fa7	P	concentrate on work rather than daydreams
NEO	Fantasy	Fantasy	Fa8	P	rarely enjoy make believe
NEO	Feelings	Non-traditionalism	Fe1	P	don't pay attention to feelings
NEO	Feelings	Non-traditionalism	Fe2	I	don't notice feelings environments produce
NEO	Feelings	Openness to Emotions	Fe3	I	wide range of Openness to Emotions
NEO	Feelings	Openness to Emotions	Fe4	I	few strong Openness to Emotions
NEO	Feelings	Openness to Emotions	Fe5	I	scents or places evoke strong moods
NEO	Feelings	Openness to Emotions	Fe6	P	strong Openness to Emotions make life interesting
NEO	Feelings	Openness to Emotions	Fe7	P	easy to empathize

NEO	Feelings	Openness to Emotions	Fe8	P	feelings are important to me
NEO	Ideas	Intellectual Interests	Id1	C	enjoy playing with abstract ideas
NEO	Ideas	Intellectual Interests	Id2	C	philosophy is boring
NEO	Ideas	Intellectual Interests	Id3	C	lose interest when talking about abstract/theoretical matters
NEO	Ideas	Intellectual Curiosity	Id4	I	enjoy "mind twister" puzzles
NEO	Ideas	Intellectual Curiosity	Id5	I	like solving problems/puzzles
NEO	Ideas	Intellectual Curiosity	Id6	C	a lot of curiosity
NEO	Ideas	Intellectual Curiosity	Id7	C	broad range of intellectual interests
NEO	Ideas	Intellectual Interests	Id8	C	no interest in thinking about the human condition
NEO	Aesthetics	Aesthetic Appreciation	NA1	C	not interested in the arts
NEO	Aesthetics	Aesthetic Appreciation	NA2	C	poetry does not effect me
NEO	Aesthetics	Aesthetic Appreciation	NA3	C	get a chill from poetry or a work of art
NEO	Aesthetics	Aesthetic Appreciation	NA4	I	ballet is boring
NEO	Aesthetics	Aesthetic Appreciation	NA5	C	interested in patterns found in art and nature
NEO	Aesthetics	Aesthetic Appreciation	NA6	P	enjoy poetry with feelings rather than images/story lines
NEO	Aesthetics	Openness to Emotions	NA7	I	get absorbed into music
NEO	Aesthetics	Openness to Emotions	NA8	I	genres of music are fascinating to me
NEO	Values	Non-traditionalism	Va1	P	religious authorities should make decisions on moral issues
NEO	Values	Non-traditionalism	Va2	P	people should know what they're doing by 25
NEO	Values	Non-traditionalism	Va3	C	controversial speakers will only confuse people
NEO	Values	Diversity	Va4	I	law changes should reflect the needs of the changing world

NEO	Values	Diversity	Va5	P	broad-minded of other's lifestyles
NEO	Values	Diversity	Va6	P	right and wrong differ for people
NEO	Values	Variety-Seeking	Va7	P	better to stick to principles than being open-minded
NEO	Values	Non-traditionalism	Va8	I	honor traditional values
Woo et al.	Curiosity	Intellectual Interests	Cu1	P	I have never really been interested in science
Woo et al.	Curiosity	Intellectual Curiosity	Cu2	I	I like to analyze things instead of taking them at face value
Woo et al.	Curiosity	Intellectual Curiosity	Cu3	I	I love to do experiments and see the results
Woo et al.	Curiosity	Intellectual Curiosity	Cu4	C	I continually strive to uncover information about topics that are new to me
Woo et al.	Curiosity	Intellectual Curiosity	Cu5	I	I try to learn something new every day
Woo et al.	Curiosity	Non-traditionalism	Cu6	P	I don't like trying new things and would rather stick with what I know
Woo et al.	Curiosity	Non-traditionalism	Cu7	C	I have no interest in learning new information
Woo et al.	Curiosity	Non-traditionalism	Cu8	I	I seldom seek new opportunities to extend my knowledge
Woo et al.	Curiosity	Diversity	Cu9	P	In a quiz I like to know what the answers are if I get the questions wrong
Woo et al.	Depth	Intellectual Interests	Dp1	C	Sometimes I avoid getting involved in philosophical discussions
Woo et al.	Depth	Intellectual Interests	Dp2	I	I'm happiest when conversations are practical rather than philosophical
Woo et al.	Depth	Intellectual Interests	Dp3	C	I am always interested in learning more about philosophy
Woo et al.	Depth	Intellectual Curiosity	Dp4	I	I take the time to reflect on my own thoughts and actions
Woo et al.	Depth	Aesthetic Appreciation	Dp5	P	For me personal growth is more important than success
Woo et al.	Depth	Intellectual Curiosity	Dp6	I	For me, there is nothing better than taking the time to think deeply about something
Woo et al.	Depth	Aesthetic Appreciation	Dp7	P	I am fascinated by meditation and processes which encourage one to look inward
Woo et al.	Depth	Non-traditionalism	Dp8	C	I believe in-depth discussions are a complete waste of time

Woo et al.	Depth	Intellectual Interests	Dp9	C	I regard philosophy as a disease of the idle
Woo et al.	Intellectual Efficiency	Self-Assessed Intelligence	Ie1	C	Tasks that require a lot of thinking confuse me easily
Woo et al.	Intellectual Efficiency	Self-Assessed Intelligence	Ie2	I	I am a slow learner
Woo et al.	Intellectual Efficiency	Self-Assessed Intelligence	Ie3	P	I always have difficulty applying new concepts
Woo et al.	Intellectual Efficiency	Self-Assessed Intelligence	Ie4	P	I often need people to explain things to me
Woo et al.	Intellectual Efficiency	Self-Assessed Intelligence	Ie5	C	I am usually not very quick in my thinking but have strengths in other areas
Woo et al.	Intellectual Efficiency	Self-Assessed Intelligence	Ie6	P	I have to read complex information several times before I fully understand it
Woo et al.	Intellectual Efficiency	Self-Assessed Intelligence	Ie7	P	I need things explained to me only once
Woo et al.	Intellectual Efficiency	Self-Assessed Intelligence	Ie8	C	I am very quick at processing information
Woo et al.	Intellectual Efficiency	Self-Assessed Intelligence	Ie9	P	I grasp scientific theories easily
Woo et al.	Ingenuity	Self-Assessed Intelligence	Ig1	I	I'm hopeless with inventing new things
Woo et al.	Ingenuity	Self-Assessed Intelligence	Ig2	P	I rarely take an idea and apply it in a new way
Woo et al.	Ingenuity	Self-Assessed Intelligence	Ig3	C	I avoid situations where I might have to come up with something new
Woo et al.	Ingenuity	Imaginative	Ig4	I	Compared to other people I don't think I am very creative
Woo et al.	Ingenuity	Intellectual Curiosity	Ig5	P	I improvise if I don't have the right tool for a job
Woo et al.	Ingenuity	Imaginative	Ig6	C	I like coming up with imaginative plans
Woo et al.	Ingenuity	Imaginative	Ig7	C	I would rather have a job that involves creativity than one that doesn't
Woo et al.	Ingenuity	Imaginative	Ig8	I	I can develop inventive ideas of high quality
Woo et al.	Ingenuity	Imaginative	Ig9	P	People come to me if they are stuck for fresh ideas
Woo et al.	Tolerance	Intellectual Interests	To1	I	I like to hear different people's views on political issues

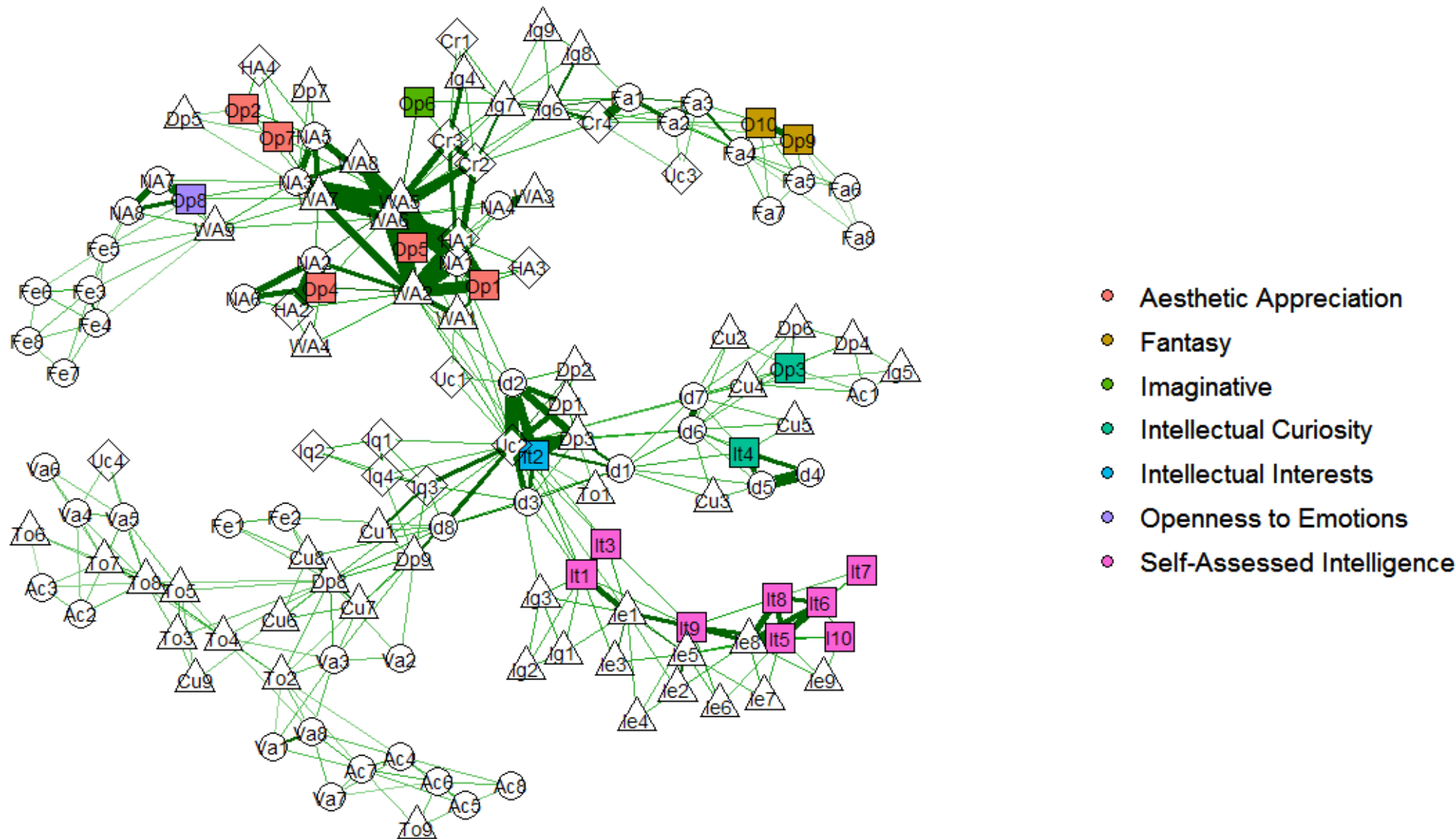
Woo et al.	Tolerance	Non-traditionalism	To2	C	I think it is rude when others speak in a language I can't understand
Woo et al.	Tolerance	Diversity	To3	I	I understand that people can have different attitudes toward certain things than I do
Woo et al.	Tolerance	Non-traditionalism	To4	I	Immigrants really irritate me
Woo et al.	Tolerance	Diversity	To5	I	Like most people, I am open to listening to what others have to say
Woo et al.	Tolerance	Diversity	To6	P	I enjoy experiencing the rituals associated with different religions
Woo et al.	Tolerance	Diversity	To7	I	I learn a great deal from people with differing beliefs
Woo et al.	Tolerance	Diversity	To8	C	I enjoy diversity in the community
Woo et al.	Tolerance	Variety-Seeking	To9	P	I prefer to visit countries where they speak my language
Woo et al.	Aesthetics	Aesthetic Appreciation	WA1	I	I think viewing art is a waste of time
Woo et al.	Aesthetics	Aesthetic Appreciation	WA2	C	Art bores me
Woo et al.	Aesthetics	Aesthetic Appreciation	WA3	P	I don't find Classical Ballet interesting
Woo et al.	Aesthetics	Aesthetic Appreciation	WA4	I	I don't find literature especially interesting
Woo et al.	Aesthetics	Aesthetic Appreciation	WA5	C	I have a passion for art
Woo et al.	Aesthetics	Aesthetic Appreciation	WA6	C	I enjoy art exhibitions
Woo et al.	Aesthetics	Aesthetic Appreciation	WA7	C	I see the beauty in art when others do not
Woo et al.	Aesthetics	Aesthetic Appreciation	WA8	I	If I see artwork I like in a gallery, I will visit it more than once to fully appreciate it
Woo et al.	Aesthetics	Openness to Emotions	WA9	C	I have been touched emotionally by a great musical performance

Note. For the NEO, the items are reworded phrases that capture the sense of the item, not the exact items from the NEO-PI-3

inventory.

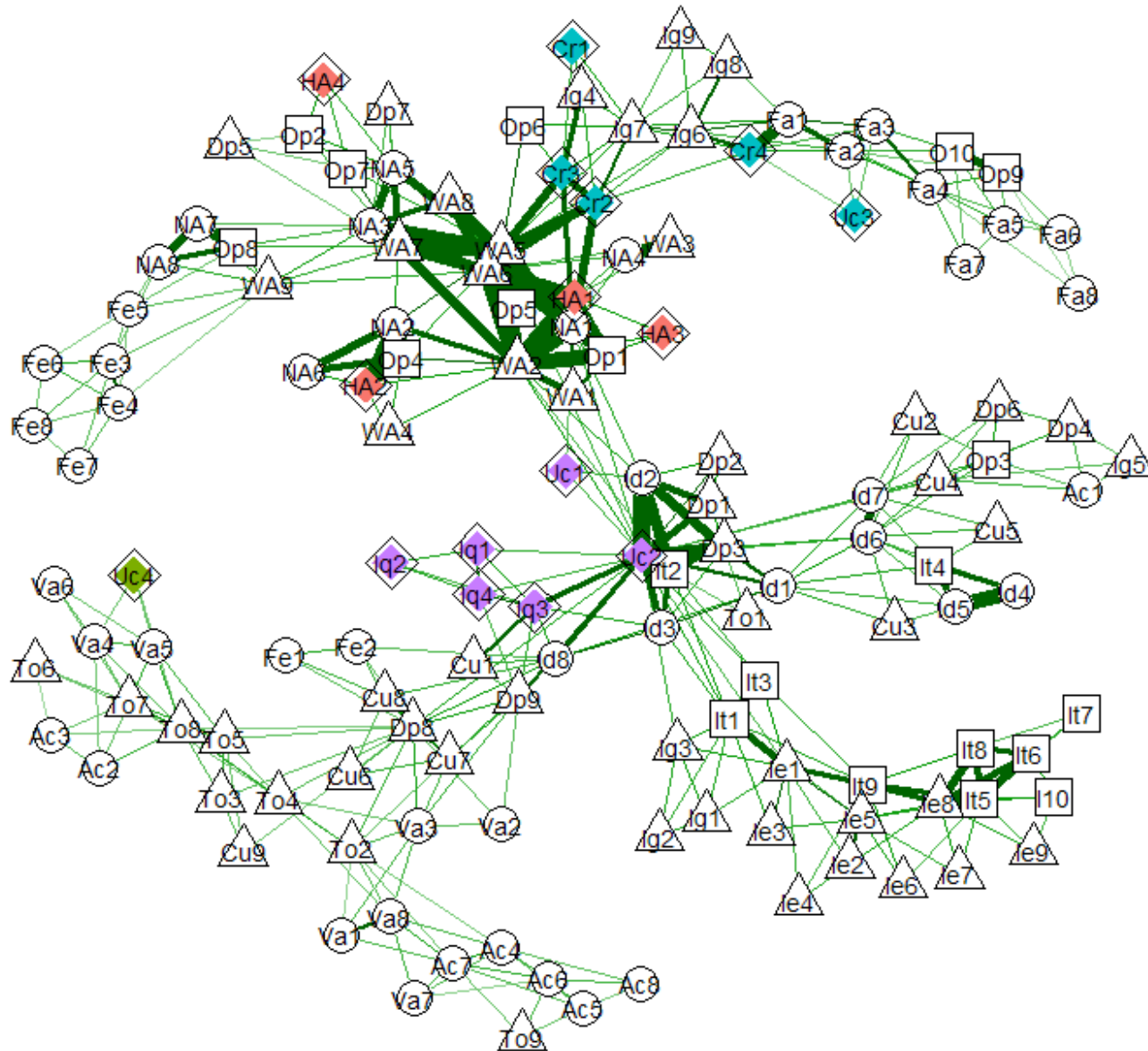
SI 3. Larger view of BFAS inventory conceptual coverage (top-left, Figure 2; see SI 1 for item descriptions)

BFAS



SI 4. Larger view of HEXACO-100 inventory conceptual coverage (top-right, Figure 2; see SI 1 for item descriptions)

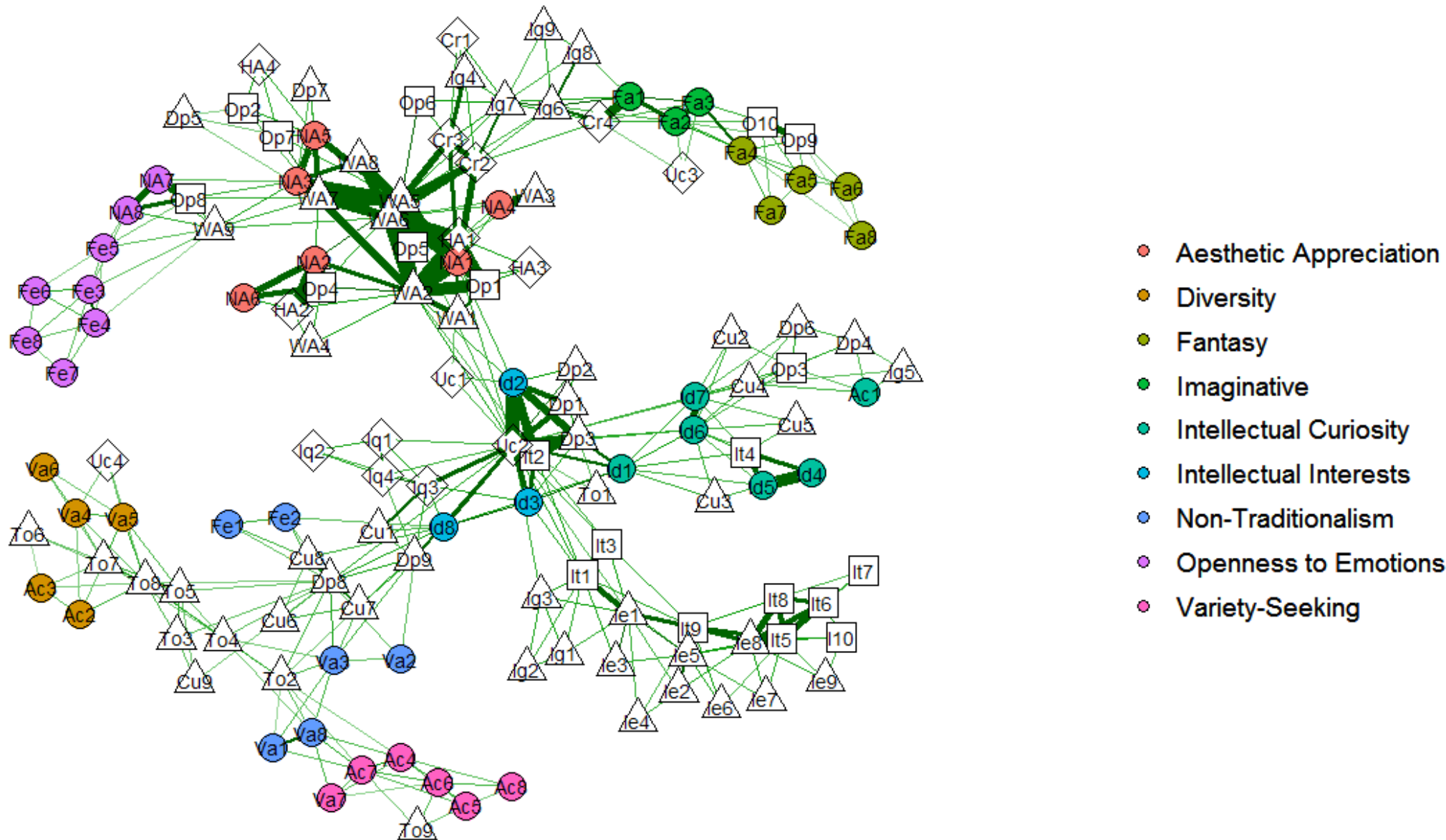
HEXACO-100



- Aesthetic Appreciation
- Diversity
- Imaginative
- Intellectual Interests

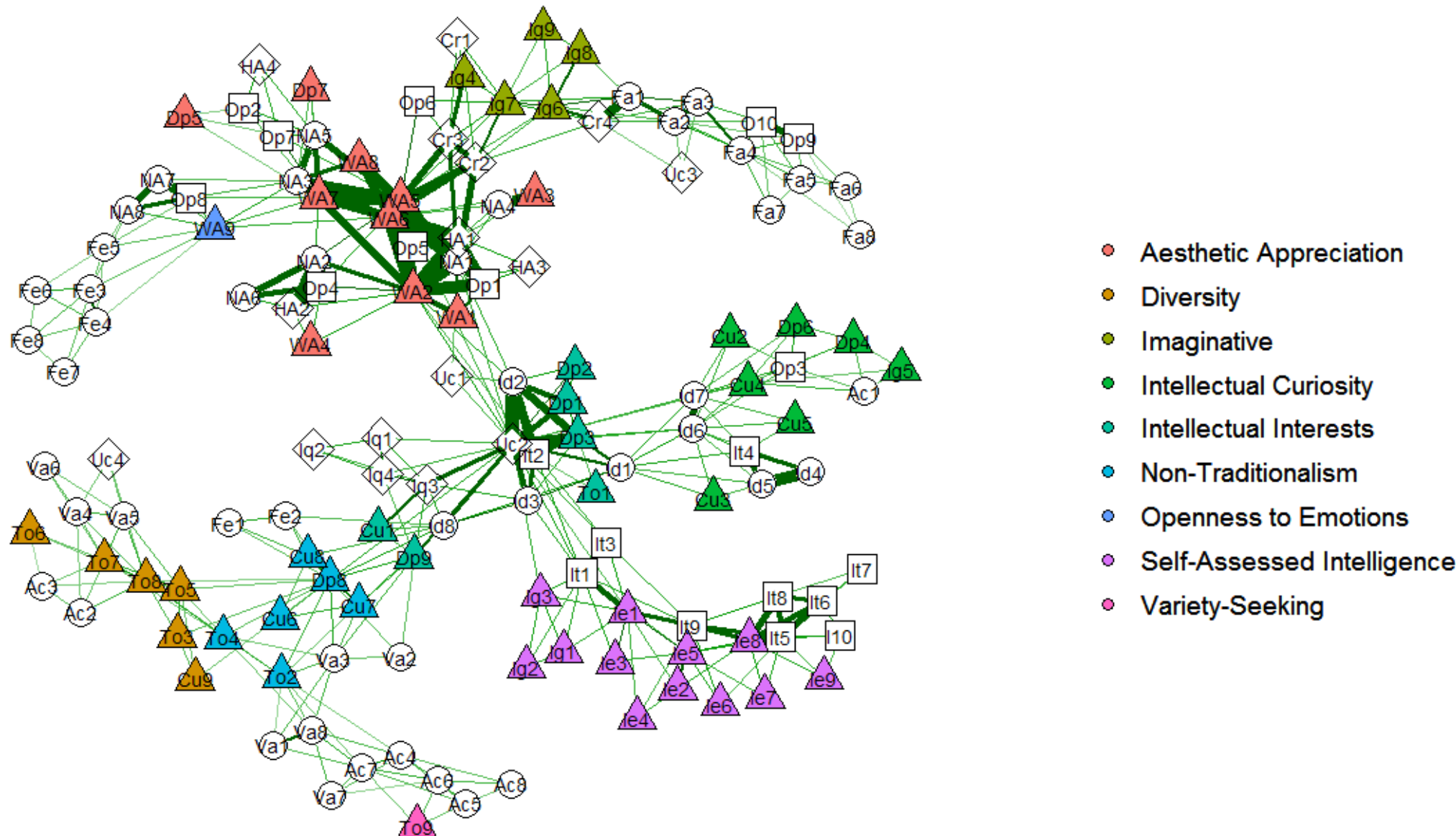
SI 5. Larger view of NEO-PI-3 inventory conceptual coverage (bottom-left, Figure 2; see SI 1 for item descriptions)

NEO-PI-3



SI 6. Larger view of Woo et al.'s (2014) Openness to Experience inventory conceptual coverage (bottom-right, Figure 2; see SI 1 for item descriptions)

Woo et al. (2014)



SI 7. MANOVA and Box's M tests for analysis of group differences and equal covariances

Statistic	MANOVA				Box's M
	Pillai	F (df)	Pillai	F (df)	F (df)
Sample		1		2	1
1 (<i>n</i> = 176)	—	—	—	—	—
2 (<i>n</i> =108)	.650	1.95 (1, 282)	—	—	—
3 (<i>n</i> = 518)	.675	8.34 (1, 692)	.614	5.62 (1, 624)	1.292 (9591)

Note: MANOVA with all three samples, $F_{(2, 799)} = 4.43$, Pillai = .960, $p < .001$. All p 's $< .001^{***}$.

SI 8. R code for all analyses in the manuscript

```
###Reopening Openness to Experience:

###A Network Analysis of Four Openness to Experience Inventories


###install necessary packages

devtools::install_github("AlexChristensen/NetworkToolbox")

install.packages(qgraph)

install.packages(igraph)


###load packages

library(NetworkToolbox)

library(qgraph)

library(igraph)


###read in data

data <- read.csv(file.choose(),header=TRUE,sep=",")


###construct network

tmfg <- TMFG(data)$A


###perform walktrap community detection algorithm

walktrap <- convert2igraph(tmfg)


#check walktrap robustness

rand<-.Random.seed

walkcheck<-matrix(0,nrow=138,ncol=10)
```

```
for(i in 1:10)
{
    set.seed(rand[i])
    walkcheck[,i]<-walktrap.community(walktrap)$membership
}
```

```
set.seed(0)
wc<-walktrap.community(walktrap)
wc$membership
max(wc$membership)
```

```
###name communities
```

```
for (i in 1:length(wc$membership))
    if (wc$membership[i] == 1)
        {wc$membership[i]<-c("Intellectual Curiosity")}
    }else if (wc$membership[i] == 2)
        {wc$membership[i]<-c("Aesthetic Appreciation")}
    }else if (wc$membership[i] == 3)
        {wc$membership[i]<-c("Self-Assessed Intelligence")}
    }else if (wc$membership[i] == 4)
        {wc$membership[i]<-c("Openness to Emotions")}
    }else if (wc$membership[i] == 5)
        {wc$membership[i]<-c("Intellectual Interests")}
    }else if (wc$membership[i] == 6)
        {wc$membership[i]<-c("Imaginative")}
```



```

    }else if (wc$membership[i] == 7)

    {wc$membership[i]<-c("Non-Traditionalism")}

    }else if (wc$membership[i] == 8)

    {wc$membership[i]<-c("Diversity")}

    }else if (wc$membership[i] == 9)

    {wc$membership[i]<-c("Variety-Seeking")}

    }else if (wc$membership[i] == 10)

    {wc$membership[i]<-c("Fantasy")}

###Full Network

layout(matrix(c(1,1)))

shapes<-

c(rep("circle",48),rep("square",20),rep("diamond",16),rep("triangle",5

4))

plot.ega <- qgraph(tmfg, label.prop=1.5, layout = "spring", vsize =

2.5, esize = 10, groups =

as.factor(wc$membership),GLratio=1.5,shape=shapes,palette="ggplot2")

####Layout the four Openness to Experience inventories

layout(matrix(c(1,3,2,4),nrow=2))

#BFAS Network

BFAS<-as.factor(c(rep(NA,48),wc$membership[49:68]))

plot.egab <- qgraph(tmfg, title = "BFAS", groups = BFAS, layout =

"spring", vsize = 4, esize =

10,GLratio=1.5,labels=FALSE,shape=shapes,palette="ggplot2")

```

```
#HEXACO Network
```

```
HEX<-as.factor(c(rep(NA,68),wc$membership[69:84]))
```

```
plot.egah <- qgraph(tmfg, title = "HEXACO-100", groups = HEX, layout =  
"spring", vsize = 4, esize =  
10, GLratio=1.5, labels=FALSE, shape=shapes, palette="ggplot2")
```

```
#NEO Network
```

```
NEO<-as.factor(wc$membership[1:48])
```

```
plot.egan <- qgraph(tmfg, title = "NEO-PI-3", groups = NEO, layout =  
"spring", vsize = 4, esize =  
10, GLratio=1.5, labels=FALSE, shape=shapes, palette="ggplot2")
```

```
#Woo Network
```

```
WOO<-as.factor(c(rep(NA,84),wc$membership[85:138]))
```

```
plot.egaw <- qgraph(tmfg, title = "Woo et al. (2014)", groups = WOO,  
layout = "spring", vsize = 4, esize = 10,  
GLratio=1.5, labels=FALSE, shape=shapes, palette="ggplot2")
```

```
#Supplementary
```

```
#BFAS Network
```

```
layout(matrix(c(1,1)))
```

```
BFAS<-as.factor(c(rep(NA,48),wc$membership[49:68]))
```

```
plot.egab <- qgraph(tmfg, title="BFAS", label.prop=1.5, groups = BFAS,  
layout = "spring", vsize = 2.5, esize =
```

```

10, GLratio=1.5, shape=shapes, palette="ggplot2")

#HEXACO Network
HEX<-as.factor(c(rep(NA,68),wc$membership[69:84]))
plot.egah <- qgraph(tmfg, title="HEXACO-100",label.prop=1.5,groups =
HEX, layout = "spring", vsize = 2.5, esize =
10, GLratio=1.5, shape=shapes, palette="ggplot2")

#NEO Network
NEO<-as.factor(wc$membership[1:48])
plot.egan <- qgraph(tmfg, title="NEO-PI-3",label.prop=1.5,groups =
NEO, layout = "spring", vsize = 2.5, esize =
10, GLratio=1.5, shape=shapes, palette="ggplot2")

#Woo Network
WOO<-as.factor(c(rep(NA,84),wc$membership[85:138]))
plot.egaw <- qgraph(tmfg, title="Woo et al.
(2014)",label.prop=1.5,groups = WOO, layout = "spring", vsize = 2.5,
esize = 10, GLratio=1.5, shape=shapes, palette="ggplot2")

```